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Propensity score matching analysis of the association between physical activity and multimorbidity in middle-aged and elderly Chinese

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In the context of an aging population, older adults increasingly face the challenge of managing multiple chronic conditions simultaneously. This study utilized analytical methods such as propensity score matching (PSM) and multivariate logistic regression, to explore the relationship between physical activity and the number of chronic diseases as well as the risk of developing co-morbidities among middle-aged and elderly Chinese individuals using data from the 2020 China Health and Retirement Longitudinal Survey. The PSM results showed that physical activity decreased the number of chronic diseases in middle-aged and elderly people by 0.050 ($p < 0.05$). The multivariate logistic regression results the odds ratio (OR) for the risk of multimorbidity in the moderate and high intensity physical activity groups compared to the group with inadequate physical activity were 0.845 (95% CI 0.729–0.980) and 0.847 (95% CI 0.727–0.988), which means that moderate-intensity physical activity is strongly associated with a reduced risk of multimorbidity. Regular physical activity among middle-aged and older adults is associated with a reduction in the number of chronic diseases they suffer from.

Keywords Multimorbidity, Co-morbidity, Chronic disease, Physical activity, PSM, CHARLS

As the fertility rate continues to decline and life expectancy steadily increases, the process of population aging is noticeably accelerating. According to the seventh China Population Census, by 2020, the number of people aged 60 and above in China had reached 264.02 million, representing 18.70% of the total population. It is projected that by 2035, China will enter a highly aging society, with the proportion of the elderly population expected to exceed 20%[1]. With the growth of the senior population, chronic diseases are becoming a major public health problem threatening the health of the older population, and multimorbidity (The presence of two or more chronic diseases in the same individual.) is becoming increasingly common among them. Specifically, the prevalence of multimorbidity in China's middle-aged and elderly population has reached 57.3% in 2018, with a trend of continuous growth, and with cardiovascular-metabolic disease multimorbidity being the most common[2,3].

Given that chronic diseases and multimorbidity are challenging and time-consuming to treat due to the complexity of the factors that influence them, more and more scholars are focusing their attention on studying the effects of physical activity on chronic diseases or multimorbidity[4]. Although some progress has been made in current research, overall the field still has the following shortcomings: Studies on the relationship between physical activity and chronic diseases or multimorbidity are primarily based on data and opinions from developed Western countries, while relatively few studies have been conducted in China and other developing countries[5]. Even when there are studies relevant to China, they tend to focus more on the relationship between physical activity and chronic diseases, with little attention paid to the relationship between physical activity and multimorbidity[6,7]. Therefore, the present study raises a topic that needs to be explored: Whether physical activity is associated with a reduction in the number of chronic diseases among middle-aged and elderly Chinese? If that is the case, then we need to consider whether there is a selection bias for physical activity in different groups of middle-aged and elderly people, and whether exercise can reduce multimorbidity coexistence in middle-aged and elderly people after discharging the selection bias? If so, do different physical activity intensities also relate to varying risks of multimorbidity?

With the above in mind, we proposed the hypothesis as follows:

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Hypothesis 1 Physical activity is associated with a reduced number of chronic diseases among middle-aged and older adults. This hypothesis is supported by a significant body of research, including a study that analyzed data from 26 different chronic diseases, which indicates that exercise can serve as a prescription for treating a variety of chronic conditions[5,8–10].

Hypothesis 2 The number of chronic diseases that can still be associated with a reduction through physical activity participation among middle-aged and older adults after controlling for selection bias. Different groups of middle-aged and older adults exhibit a selection bias towards physical activity, and we employed propensity score matching (PSM) to address endogeneity and demonstrate the robustness of our results. If, after controlling for this bias, physical activity is still linked to a decrease in the number of chronic diseases, it underscores its significant health benefits[11,12].

Hypothesis 3 The association between varying intensities of physical activity and the prevalence of multimorbidity among middle-aged and elderly individuals varies. An extensive body of literature indicates that physical activity can reduce the prevalence of multimorbidity and enhance the quality of life for this demographic[13–16]. However, there are fewer studies on the relationship between different intensities of physical activity and the prevalence of multimorbidity in Chinese middle-aged and older adults.

Methods
Data
For the 2020 China Health and Retirement Longitudinal Survey (CHARLS) database, we paid special attention to and selected a specific group of study participants, the middle-aged and older population. This group, clearly defined as those aged 45 and over, represents a significant portion of society and is the key focus of our study. The CHARLS is a large-scale interdisciplinary survey project hosted by the National Development Research Institute of Peking University. The project was approved by the Ethics Committee of Peking University (IRB00001052-11015). The CHARLS database is a high-quality, public micro-database that includes detailed socio-economic and health information. It can provide a more comprehensive and accurate description of the life and health status of Chinese residents aged 45 years and above, as well as the context of their family and social relationships[17]. The inclusion criteria for this study were as follows: participants must be aged 45 years or above and have a household registration. The exclusion criteria included any instance where a variable was missing, ambiguous, or did not meet the requirements of the study. A valid sample consisting of 10,118 individuals was obtained (See Fig. 1).

Variables
Outcome variables
The dependent variable is the number of chronic diseases, and CHARLS investigates a total of 15 chronic diseases, including hypertension, dyslipidemia, diabetes or elevated blood glucose, malignant neoplasms such as cancer, chronic lung disorders, liver disorders, heart disease, stroke, kidney disorders, gastric disorders or disorders of the digestive system, emotional and psychiatric problems, memory-related disorders, Parkinson's disease, arthritis or rheumatic disorders, and asthma. Chronic diseases are recorded on the basis of patient self-reporting of confirmed diagnosis. The total number of chronic diseases suffered by the respondents was counted and categorized into five groups: 0 (those with no chronic diseases), 1, 2, 3 and ≥ 4 (those with four or more chronic diseases). We also defined specifies respondents with two or more chronic diseases are considered to have multimorbidity, while those with one or no chronic disease are considered to have no multimorbidity[16].

Core explanatory variable
The core independent variable is "physical activity", i.e., whether middle-aged and older adults are physically active. We use the MET-mins calculation, we assign a value of 3.3 METs to light-intensity physical activities (such as walking), 4 METs to moderate-intensity physical activities, and 8 METs to vigorous-intensity activities that are highly exhausting, Finally, the respondents' physical activity days per week, physical activity time per day, and MET values were multiplied to obtain the total weekly physical activity level. The subjects were also divided into 3 groups according to their total weekly physical activity level: High Intensity Physical Activity Group (high intensity physical activity on a combined total of no less than 7 days per week with a combined total energy expenditure of 1500 MET-mins; high intensity physical activity on at least 3 days per week with a cumulative total of at least 1500 MET-mins per week; or walking and engaging in moderate-intensity or vigorous-intensity physical activity every day with a cumulative total of 3000 MET-mins per week)、Moderate Intensity Physical Activity Group (high-intensity physical activity for at least 20 min per day, at least 3 days per week; moderate-intensity physical activity or walking for at least 30 min per day, at least 5 days per week; or walking and participating in moderate-intensity or high-intensity physical activity for at least 5 days per week, with a cumulative total of 600 MET-mins per week) and Group with Inadequate Physical Activity (not at the intensity of either of the above or less than 10 min of exercise per week)[18]. We also define the group with inadequate physical activity as physically inactive, and all others as physically active.

Control variables
The control variables were selected in terms of socio-demographic characteristics (age, gender, residential address), social support (marital status, social support, level of education), health behaviors (smoking, drinking, sleeping), and socio-economic status (whether or not they receive a retirement pension, and how they pay for their medical expenses).

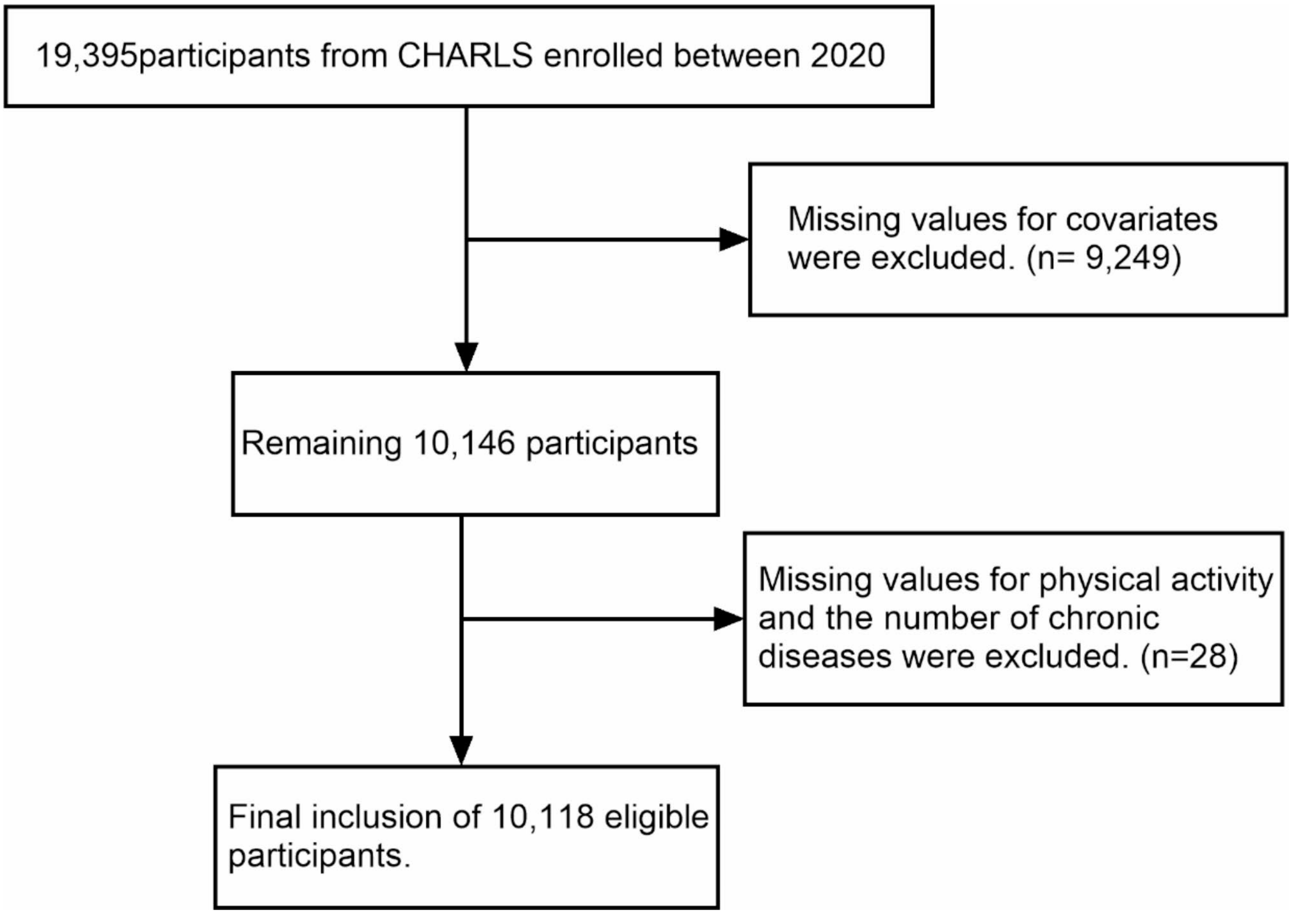

Fig. 1. Flowchart for Participant Screening and Exclusion Criteria.

Statistical analysis

Basic analysis

SPSS 26.0 and Stata 17.0 software were used to complete the statistical analysis. The Chi-square test was used for descriptive statistics to observe the distribution of data. The regression analysis section utilizes both zero-inflated negative binomial regression (ZINB) models and ordinary least squares (OLS) models to estimate the relationship between the variables. In addition, propensity score matching was used to address respondents' selection bias. Finally, logistic regression-derived odds ratio (OR) was utilized to investigate the association between varying exercise intensities and the likelihood of multimorbidity.

Propensity score matching

The engagement in physical activity by middle-aged and older adults constitutes a self-selective behavior, given that physical activity participation is not mandated, thereby introducing an element of endogeneity. If not effectively mitigated, the influence of physical activity on the prevalence of chronic diseases may lack internal validity in the regression outcomes, potentially leading to biased findings. PSM can overcome the endogenous threat posed by self-selection and solve the problem of selection bias. The average treatment effect (ATT) for the participants was derived through the modeling of the following:

$ATT = E\,[Y_{1i}\,|D_i = 1, p(X_i)] - E\,[Y_{0i}|\,D_i = 0, p(X_i)]$.

In the construction of a propensity score matching (PSM) model, the initial step involves fitting a model to estimate the probability of each individual receiving treatment, such as engaging in physical activity. This probability is referred to as the "propensity score." PSM utilizes a binary logistic regression model to fit the data, which is used to predict the relationship between a binary response variable and explanatory variables. In this context, the response variable Di indicates whether an individual engages in physical exercise, where Di = 1 signifies elderly individuals who exercise and Di = 0 denotes those who do not. The explanatory variables include age, gender, health status, socioeconomic status, and any other factors that might influence treatment selection. Within the logistic regression model, we use the covariates Xi to predict the probability of Di = 1, denoted as P(Di = 1|Xi). This probability, known as the propensity score, represents the likelihood of being assigned to the treatment group (elderly individuals who physical activity) given the covariates. Once the model is fitted, we calculate these scores for all individuals and use them for matching to ensure similar covariate distributions between treatment and control groups. Various matching algorithms, such as nearest neighbor or radius matching, can be employed. By comparing the outcomes of matched groups, we estimate the average treatment effect (ATT). In this study, the ATT will inform us about the change in the number of chronic diseases among elderly individuals who physical activity if they were not to engage in physical activity. Thus, observed differences are likely due to the treatment effect, minimizing confounding factors[11]. Ultimately, the assessment of whether the matching results are balanced or not is based on two criteria: examining the kernel density map and ensuring that the deviation rate is less than 10%[19,20].

Results

Characteristics of the study sample

Table 1 presents the intergroup disparities between each categorical variable and the number of chronic diseases. The analyses revealed a statistically significant difference in physical activity engagement among middle-aged and older adults when stratified by the number of chronic diseases prevalent. In addition, descriptive statistics were performed on 15 chronic diseases (see Table S1). The results indicated that dyslipidemia was the most prevalent chronic disease among all participants and across different strata based on the number of chronic diseases. Parkinson's disease exhibited the lowest prevalence among all participants and those with 1 to 3 concurrent chronic diseases. However, for participants with four or more concurrent chronic diseases, cancer had the lowest incidence rate at 5.2%.

Basic results analysis

The dependent variables in this study are quantitative variables that cannot be taken as negative or decimal. Consequently, we are considering the application of either Poisson regression or negative binomial regression to investigate the correlation between the independent variables and the dependent variable. Ultimately, the ZINB model was chosen because the dependent variable was too dispersed and more than 50% were zero. It also utilizes the OLS model to augment the robustness of the estimated results. Table 2 reports the estimation results after adding different control variables, in which Models (1) and (2) are the ZINB regression results, and Models (3) and (4) are the OLS regression results. The Models (1) and (3) do not include any control variables, and Models (2) and (4) add variables for individual characteristics, family characteristics, and social characteristics. After controlling for individual, family, and social factors, Model 2 from the regression analysis indicated that middle-aged and older adults who engaged in physical activity had a significantly lower number of chronic disease conditions by 0.095 ($p < 0.01$), compared to their inactive counterparts. These results were in line with Hypothesis 1.

In addition to the protective effect of physical activity, the regression analysis identified several confounding variables that significantly influence the number of chronic diseases among middle-aged and older adults. Factors such as age, sleep, marital status and alcohol consumption, etc. Specifically, middle-aged people have fewer chronic diseases than older people; middle-aged and older people who do not get enough sleep have more chronic diseases than those who do not; middle-aged and older people in stable marriages have fewer diseases than those who are divorced or widowed; and middle-aged and older people who do not consume alcohol have fewer chronic diseases than those who consume alcohol.

PSM estimation results

Figure 2 presents the distribution of propensity scores for both treatment and control groups, before and after the application of Propensity Score Matching (PSM). The overlap of the two curves in the graph indicates that, prior to PSM, there was a significant difference in propensity scores between the treatment and control groups, suggesting substantial variation in the observed covariates. However, after applying PSM, this disparity was greatly diminished. The convergence in the conditional probability distributions across the two groups suggests that the matching procedure successfully balanced the measured covariates, thereby mitigating selection bias. This improved comparability between the groups enhances the reliability of subsequent comparisons related to the outcome of interest, facilitating more robust conclusions about the efficacy of the physical activity intervention.

Moreover, a balance test was conducted, and the results are shown in Table 3. The deviation rate of all variables after matching was reduced to < 10%, and the differences between all variables were no longer significant after checking, which also showed that the matching effect was great.

To compare the treated and control groups, we utilized 1:1 Individual Matching, K-Nearest Neighbor Matching, Caliper Matching, and Kernel Matching, and subsequently calculated the treatment effect. The results of the estimation using the PSM method are presented in Table 4. Based on the average of the ATT values obtained from the four matching techniques, it was concluded that physical activity can reduce the number of chronic diseases among middle-aged and older adults by 0.050. This is consistent with hypothesis 2 that there is variability in whether or not different groups are physically active, but after addressing selection bias, physical activity still reduces the number of chronic diseases in middle-aged and older adults.

We conducted additional sensitivity analysis on the robustness of the main findings. In the regression and propensity score matching model, six chronic diseases (diabetes, heart disease, emotional and mental problems, arthritis or rheumatism, hypertension and dyslipidemia) with a high prevalence rate of middle-aged and elderly people in this study were added as confounding factors, and the results obtained did not change substantially, as shown in Table S3, S4 and S5.

Analysis of the association between different exercise intensities and the risk of multimorbidity in middle-aged and elderly people in China

In order to explore the analysis of the association between different physical activity intensities and the risk of developing multimorbidity in middle-aged and elderly people, the following regression analyses were performed. We categorized respondents into two groups based on whether or not they had multimorbidity, (0: Without

Variables	Number of chronic diseases (n: Sample size)					X^2	P
	0($n=6\,416$)	1($n=2\,430$)	2($n=853$)	3($n=265$)	≥4($n=154$)		
Physical activity (%)							
No	1 666(26.0)	616(25.3)	239(28.0)	83(31.3)	63(40.9)	23.152	< 0.001
Yes	4 750(74.0)	1 814(74.7)	614(72.0)	182(68.7)	91(59.1)		
Age (%)							
Middle age (45 ~ 59)	2 994(46.7)	946(38.9)	286(33.5)	101(38.1)	44(28.6)	98.182	< 0.001
Old age (≥ 60)	3 422(53.3)	1 484(61.1)	567(66.5)	164(61.9)	110(71.4)		
Gender (%)							
Male	1 140(17.8)	414(17.0)	136(15.9)	57(21.5)	28(18.2)	5.089	0.278
Female	5 276(82.2)	2 016(83.0)	717(84.1)	208(78.5)	126(81.8)		
Residential address (%)							
Towns and cities	2 285(35.6)	900(37.0)	289(33.9)	112(42.3)	49(31.8)	8.878	0.064
Countryside	4 131(64.4)	1 530(63.0)	564(66.1)	153(57.7)	105(68.2)		
Marital (%)							
status Normal	4 858(75.7)	1 764(72.6)	628(73.6)	187(70.6)	102(66.2)	17.700	< 0.001
Other	1 558(24.3)	666(27.4)	225(26.4)	78(29.4)	52(33.8)		
Social activities (%)							
Yes	3 040(47.4)	1 211(49.8)	411(48.2)	138(52.1)	76(49.4)	6.000	0.199
No	3 376(52.6)	1 219(50.2)	442(51.8)	127(47.9)	78(50.6)		
Education (%)							
Primary school or below	4 395(68.5)	1 709(70.3)	627(73.5)	191(72.1)	118(76.6)	14.774	0.005
Secondary school or above	2 021(31.5)	721(29.7)	226(26.5)	74(27.9)	36(23.4)		
Smoking (%)							
Yes	148(2.3)	79(3.3)	26(3.0)	11(4.2)	10(6.5)	17.485	0.002
No	6 248(97.7)	2 351(96.7)	827(97.0)	254(95.8)	144(93.5)		
Drinking (%)							
Yes	1 488(23.2)	568(23.4)	161(18.9)	50(18.9)	30(19.5)	11.756	0.019
No	4 928(76.8)	1 862(76.6)	692(81.1)	215(81.1)	124(80.5)		
Sleeping (%)							
< 6 h	2 237(34.9)	966(39.8)	377(44.2)	125(47.2)	84(54.5)		
6 ~ 8 h	3 743(58.3)	1 282(52.8)	408(47.8)	119(44.9)	62(40.3)	80.726	< 0.001
> 8 h	436(6.8)	182(7.5)	68(8.0)	21(7.9)	8(5.2)		
Pension (%)							
Yes	3 051(47.6)	1 287(53.0)	485(56.9)	129(48.7)	82(53.2)	40.537	< 0.001
No	3 365(52.4)	1 143(47.0)	368(43.1)	136(51.3)	72(46.8)		
Medical (%)							
payment Insurance	6 105(95.2)	2 322(95.6)	809(94.8)	249(94.0)	149(96.8)	2.647	0.618
Out of pocket	311(4.8)	108(4.4)	44(5.2)	16(6.0)	5(3.2)		

Table 1. Descriptive statistics of the variables.

	ZINB				OLS			
	(1)		(2)		(3)		(4)	
	Coefficient	S.E.	Coefficient	S.E.	Coefficient	S.E.	Coefficient	S.E.
Physical activity	-0.218***	0.028	-0.095**	0.035	-0.068***		-0.053**	0.020
Gender			-0.039	0.046			-0.018	0.025
Residential address			-0.039	0.034			-0.026	0.019
Marital status			0.081*	0.035			0.044*	0.020
Medical payment Insurance			-0.023	0.072			-0.021	0.041
Age			0.314***	0.040			0.165***	0.024
Sleeping			-0.146***	0.026			-0.083***	0.015
Social activities			-0.096**	0.032			-0.053**	0.018
Smoking			-0.311***	0.075			-0.216***	0.055
Drinking			0.104*	0.041			0.055*	0.022
Pension			0.070	0.039			0.033	0.023
Education			-0.058	0.035			-0.034	0.021
Constant	-0.637***	0.066	-1.550***	0.340	0.598***	0.017	0.917***	0.150
lnalpha	-2.296***	0.338	-1.014***	0.263				
N	10,118		10,118		10,118		10,118	
R^2					0.001		0.017	

Table 2. Results of ZINB and OLS. S.E. is the robust standard error, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

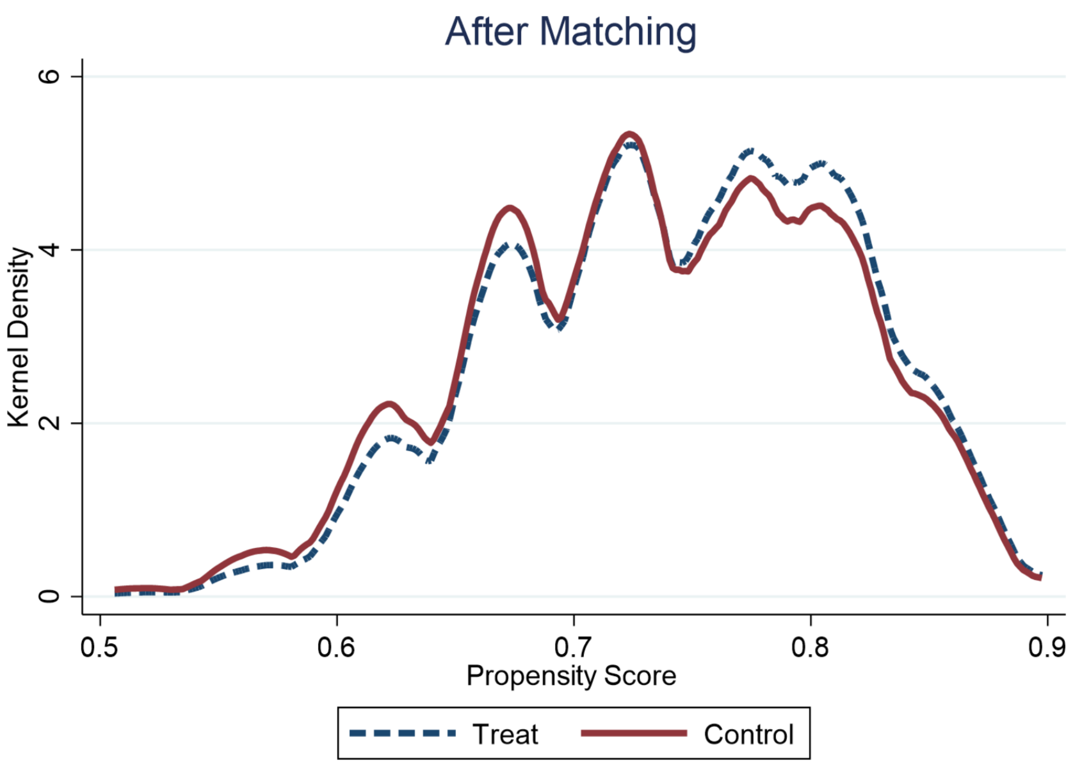

Fig. 2. Propensity distribution of the treated and control groups before and after matching.

multimorbidity, 1: Individuals with multimorbidity). Physical activity was categorized into 3 groups based on the total weekly physical activity levels of middle-aged and older adults, (0: Group with Inadequate Physical Activity, 1: Moderate Intensity Physical Activity Group, 2: High Intensity Physical Activity Group). Univariate and multivariate logistic regression analyses of physical activity were conducted sequentially, adjusting for possible confounders such as general population characteristics, level of education, and smoking status, et al. (see S2). The results showed that moderate to high intensity physical activity was negatively associated with multimorbidity after adjusting for confounders. Additionally, we conducted an association analysis between moderate and high intensity physical activity and the risk of multimorbidity, as shown in Table 5. Univariate

Variable	Unmatched / Matched	Mean Treated	Mean Control	% Bias	%Reduction |Bias|	t-test t	t-test p>|t|
Gender	U	1.821	1.835	-3.9		-1.71	0.087
	M	1.821	1.822	-0.4	89.2	-0.25	0.800
Residential address	U	1.628	1.675	-9.9		-4.34	< 0.001
	M	1.628	1.643	-3.1	68.7	-1.86	0.063
Marital status	U	1.239	1.300	-13.9		-6.29	< 0.001
	M	1.239	1.235	0.9	93.9	0.54	0.587
Medical payment Insurance	U	1.043	1.062	-8.3		-3.85	< 0.001
	M	1.043	1.038	2.2	73.7	1.51	0.131
Age	U	1.543	1.638	-19.4		-8.51	< 0.001
	M	1.543	1.554	-2.1	89.0	-1.28	0.202
Sleeping	U	1.701	1.682	3.2		1.43	0.152
	M	1.701	1.700	0.2	92.2	0.15	0.878
Social activities	U	1.481	1.622	-28.7		-12.62	< 0.001
	M	1.481	1.496	-3.1	89.3	-1.85	0.064
Smoking	U	1.972	1.976	-2.6		-1.14	0.253
	M	1.972	1.974	-1.2	55.9	-0.69	0.488
Drinking	U	1.757	1.818	-14.9		-6.45	< 0.001
	M	1.757	1.776	-4.6	69.2	-2.71	0.007
Pension	U	1.507	1.489	3.8		1.67	0.094
	M	1.507	1.506	0.2	93.4	0.15	0.879
Education	U	1.321	1.258	14.0		6.11	< 0.001
	M	1.321	1.306	3.3	76.1	1.99	0.046
Sample	Ps R^2	LR chi2	p > chi2	Mean Bias	Med Bias	B	R
Unmatched	0.025	297.07	0.000	11.1	9.9	39.0*	0.94
Matched	0.001	19.02	0.061	1.9	2.1	7.1	1.15

Table 3. Results of PSM data-balanced test. Note: * if B > 25%, R outside [0.5;2]; Ps R^2represents the Pseudo R-squared LR chi2 denotes the chi-square statistic, B represents the absolute standard deviation, and R denotes the standard deviation ratio.

Method	Treated Group	Control Group	ATT	Standard Error
1: 1 Individual Matching	0.530	0.581	-0.051**	0 0.024
K-Nearest Neighbor Matching	0.530	0.580	-0.050**	0.024
Caliper Matching	0.530	0.576	-0.046**	0.022
Kernel Matching	0.530	0.581	-0.051**	0.022

Table 4. Average treatment effect of physical activity on the number of chronic diseases among middle-aged and elderly people. Note: ** $p < 0.05$, and k = 4; ATT represents the average treatment effects on treated.

analysis results indicated that both the moderate-intensity and high-intensity physical activity groups exhibited a reduced risk of multimorbidity compared to the group with inadequate physical activity; The results of the multivariate analysis demonstrated that, after adjusting for potential confounders, the risk of multimorbidity in the moderate-intensity exercise group was 0.845 times that of the group with inadequate physical activity ($P = 0.025$); The risk of multimorbidity in the high-intensity exercise group was 0.847 times that of the group with inadequate physical activity ($P = 0.034$). This shows that different physical activity intensities have different effects on the risk of multimorbidity in middle-aged and older adults, which is consistent with Hypothesis 3.

Discussion

Chronic diseases are characterized by difficulty in curing and long duration of illness. When a patient with a chronic disease accumulates lesions over an extended period, reaching a certain level of severity, it may lead to complications, culminating in the concurrent development of multimorbidity. Jowsey et al. demonstrated that as the number of chronic diseases increases, it leads to a heightened overall economic burden of illness and a further reduction in the quality of life among middle-aged and older adults[21]. Researchers as early as 2006 have

Variable	Univariate Analysis		Multivariate Analysis	
	OR [95%CI]	*P*	OR [95%CI]	*p*
inadequate physical activity				
Moderate Intensity Physical Activity	0.812[0.703,0.938]	0.005**	0.845[0.729,0.980]	0.025*
High Intensity Physical Activity	0.788[0.678,0.915]	0.002**	0.847[0.727,0.988]	0.034*
Gender	0.987[0.845,1.152]	0.866		
Residential address	0.973[0.861,1.100]	0.663		
Medical payment Insurance	0.923[0.706,1.207]	0.560		
Pension	1.256[1.116,1.413]	< 0.001***	0.939[0.803,1.098]	0.428
Education	1.251[1.096,1.429]	0.001***	1.088[0.938,1.262]	0.267
Marital status	0.868[0.761,0.990]	0.034*	0.954[0.834,1.092]	0.495
Age	1.567[1.385,1.773]	< 0.001***	1.505[1.273,1.779]	< 0.001***
Sleeping	0.641[0.567,0.724]	< 0.001***	0.683[0.603,0.775]	< 0.001***
Social activities	1.044[0.928,1.174]	0.471		
Smoking	1.457[1.058,2.006]	0.021*	1.537[1.098,2.153]	0.012*
Drinking	0.772[0.665,0.896]	0.001***	0.778[0.663,0.913]	0.002**

Table 5. Logistic regression analysis results for the association between various physical activity intensities and the risk of multimorbidity. $^{***}P < 0.001$, $^{**}P < 0.01$, $^{*}P < 0.05$.

suggested that physical activity positively impacts the prevention of premature death among those with chronic diseases[22]. Subsequently, in 2016, a study by Jami and other scholars re-emphasized that regular physical activity is beneficial in improving physical and mental functioning and symptoms of chronic diseases in older adults, thus helping them to maintain mobility and self-care in daily life[23]. Our analysis of data from middle-aged and older adults in the CHARLS database further corroborates the idea that physical activity substantially reduces the number of chronic diseases, also confirming Hypotheses 1 and 2. This finding resonates with numerous existing studies[14,24,25]. Based on observations and analysis, we attributed the results to the following factors: Firstly, physical activity can mitigate chronic diseases such as cardiovascular, respiratory, gastrointestinal, and endocrine disorders by enhancing endothelial vasodilatation, decreasing peripheral inflammatory markers and antioxidant contingencies, among other physiological and biochemical functions[26–29]. Secondly, physical activity can also diminish the risk of depression and other mental illnesses by enhancing neurotransmitter levels, fostering the secretion of dopamine, serotonin, and other neurotransmitters in the body, thereby improving a person's mood and overall sense of well-being[30]. Therefore, based on a comprehensive assessment of the relevant literature and empirical data, we contend that physical activity operates through the mechanisms delineated above, thereby diminishing the number of chronic diseases.

This study utilized logistic regression analysis to examine the association between varying intensities of physical activity and multimorbidity occurrence. Results indicated that engagement in moderate- and high-intensity physical activities was associated with a 15.5% and 13.3% trend towards reduced multimorbidity incidence, respectively, compared to individuals with sedentary lifestyles or those under-exercising. These findings support Hypothesis 3, suggesting distinct relationships between physical activity levels and multimorbidity rates. Notably, for middle-aged and older adults, even moderate-intensity exercise significantly lowered their co-morbidity risk. This exercise intensity is lower than Yogini's findings based on a UK Biobank Resource study of people in the 38–73 age group, which concluded that weekly vigorous physical activity is recommended to minimize the effects of co-morbidities[16]. The average age of the participants selected for this study was higher than that of the subjects in the Yogini study, which may be one of the reasons for the difference. In addition, the Yogini study deals primarily with the more affluent groups in the UK, who typically have more opportunities and resources to participate in a variety of high-intensity physical activities and fitness programs. In contrast, Chinese middle-aged and older adults may receive less support and encouragement to engage in high-intensity physical activity at the community and societal levels. At the same time, the optimal intensity of physical activity derived from this study is higher than that found in Vijay's research, which suggests that low-intensity physical activity can have a positive impact on the elderly[31]. The reason for this analysis was that the study focused on the relationship between physical activity and memory impairment and dementia in older adults, without incorporating other chronic diseases. In addition, this study only found that low-intensity physical activity had a more significant effect on older women compared to our study population, which limits the generalizability and scope for generalizing the use of such exercise intensities.

Although the optimal intensity of physical activity for middle-aged and older adults varies from study to study, the results suggest that a certain level of physical activity is effective in reducing the risk of multimorbidity. This is essential for improving the quality of life of middle-aged and elderly people, indicating that middle-aged and elderly people should engage in appropriate physical activities in their daily lives to prevent the occurrence of multimorbidity.

In addition, the results of regression analysis in this study showed that age, sleep duration, and whether or not they smoked had an effect on the emergence of multimorbidity in middle-aged and older adults. Specifically, the probability of multimorbidity coexistence increased with age, which is consistent with Nazish's findings[32] in the area of aging and multimorbidity. That is, an individual's health declines progressively with age, and this decline in physiologic status further contributes to the increased incidence of chronic disease in the elderly, as well as an increased risk of developing a wide range of illnesses. Second, the risk of multimorbidity was higher among those with sleep deprivation and smoking in the middle-aged and older age groups, which is also in line with the findings of previous studies[15,33]. The reason for this is that sleep deprivation affects the body's endocrine, immune and neurological functions, leading to a wide range of chronic diseases associated with middle-aged and elderly people[34]. Smoking directly affects the respiratory system, increasing the risk of lung disease and possibly further impairing the body's physiological functions, which can lead to the coexistence of multiple diseases[35]. In addition to this, logistic regression results show that alcohol consumption reduces the risk of multimorbidity, although this is contrary to the findings of some studies[36], But there are also studies showing that light drinking, especially wine, can reduce the risk of myocardial infarction and diabetes[37]. In summary, as age rises, middle-aged and older adults should endeavor to sleep longer and quit smoking while remaining physically active as a means of preventing an increased risk of multimorbidity.

Conclusion

With the acceleration of population aging in China, the challenges posed by chronic diseases among middle-aged and elderly people are becoming increasingly severe. This study reveals that physical activity is significantly associated with a reduction in the number of chronic diseases prevalent among middle-aged and older adults, and moderate-intensity physical activity, in particular, is associated with a lower prevalence of multimorbidity in this population. These findings suggest that further research and promotion of appropriate physical activities for middle-aged and older adults may help alleviate the growing challenge of chronic disease.

Limitations and prospects

Several limitations of this study need to be further explored in future research: Firstly, although the use of PSM methods can reduce the impact of selection bias to some extent, future studies should consider implementing a longitudinal follow-up design in order to explore the causal relationship between physical activity and multimorbidity in middle aged and older adults in more depth. Secondly, in this study, participants with missing values were excluded from the analysis, potentially introducing bias. Future research should consider employing advanced statistical methods to address missing data and improve accuracy and comprehensiveness of results. Thirdly, in this study, we considered the number of 15 chronic diseases among middle-aged and elderly individuals. However, due to practical constraints and the complexity involved in data analysis, we elected to include only six of the most prevalent chronic diseases as confounders in our sensitivity analysis. This selection may not fully encompass all potential confounders that could influence the relationship between physical activity and the number of chronic diseases, thereby introducing some limitations. We encourage future researchers to consider these additional control variables in similar studies to further validate our findings. Fourthly, this study was based on questionnaire data from the CHARLS database, which may introduce subjectivity bias. Therefore, to ensure the objectivity and verifiability of the findings, future experimental studies are necessary to further confirm our findings. Finally, while the present study identified a significant association between physical activity and the risk of multimorbidity, the cross-sectional nature of the design precludes establishing a causal relationship due to the inability to observe dynamic changes between variables over time. Consequently, it is advisable for future research to employ longitudinal or experimental designs to more rigorously examine this relationship.

Data availability

Raw data for this article is available from the CHARLS website(https://charls.pku.edu.cn); The datasets used and/or analyzed during the current study are available from the first author on reasonable request.

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References

1. The results of the seventh national census. China's population is 1.41178 billion. *Xinmin Wkly.* **17**, 35 (2021).
2. Lu-wen, Z. H. A. N. G., Qiao-chu, L. U. & Yang, Z. H. A. O. Association of multimorbidity with health outcomes in the Chinese middle-aged and elderly population. *Med. Sci.*. **44** (01), 159–168 (2023).
3. Li, X. U. et al. Shifts in chronic disease and comorbidity patterns among Chinese older adults: An analysis based on the China Health and Retirement Longitudinal Study. *Chin. Gen. Pract.* **27** (11), 1296–1302. https://doi.org/10.12114/j.issn.1007-9572.2023.0634 (2024).
4. Bricca, A. et al. Benefits and harms of exercise therapy in people with multimorbidity: A systematic review and meta-analysis of randomised controlled trials. *Ageing Res. Rev.* **63**, 101166 (2020).
5. Pedersen, B. K. & Saltin, B. Exercise as medicine – Evidence for prescribing exercise as therapy in 26 different chronic diseases. *Scand. J. Med. Sci. Sports*. **25**, 1–72 (2015).
6. Zhang, H-J. et al. Effects of moderate and vigorous exercise on nonalcoholic fatty liver disease: A randomized clinical trial. *JAMA Intern. Med.* **176**, 1074–1082 (2016).
7. Li, Z. & Wang, X-Q. Clinical effect and biological mechanism of exercise for rheumatoid arthritis: A mini review. *Front. Immunol.* **13**, 1089621 (2022).
8. Luan, X. et al. Exercise as a prescription for patients with various diseases. *J. Sport Health Sci.* **8**, 422–441 (2019).
9. Anderson, E. & Durstine, J. L. Physical activity, exercise, and chronic diseases: A brief review. *Sports Med. Health Sci.* **1** (1), 3–10 (2019).
10. Mcleod, J. C., Stokes, T. & Phillips, S. M. Resistance exercise training as a primary countermeasure to age-related chronic disease. *Front. Physiol.* **10**, 645 (2019).
11. Kane, L. T. et al. Propensity score matching: a statistical method. *Clin. Spine Surg.* **33** (3), 120–122 (2020).

12. Benedetto, U. et al. Statistical primer: Propensity score matching and its alternatives. *Eur. J. Cardio-Thorac Surg. Off J. Eur. Assoc. Cardio-Thorac Surg.* **53** (6), 1112–1117 (2018).
13. Dekker, J., Buurman, B. M. & van der Leeden, M. Exercise in people with comorbidity or multimorbidity. *Health Psychol. Off J. Div. Health Psychol. Am. Psychol. Assoc.* **38** (9), 822–830 (2019).
14. Dhalwani, N. N. et al. Association between lifestyle factors and the incidence of multimorbidity in an older English population. *J. Gerontol. Biol. Sci. Med. Sci.* glw146 (2016).
15. Hanlon, P. et al. Frailty and pre-frailty in middle-aged and older adults and its association with multimorbidity and mortality: A prospective analysis of 493 737 UK Biobank participants. *Lancet Public. Health.* **3** (7), e323–e332 (2018).
16. Chudasama, Y. V. et al. Physical activity, multimorbidity, and life expectancy: A UK Biobank longitudinal study. *BMC Med.* **17** (1), 108 (2019).
17. Zhao, Y., Hu, Y., Smith, J. P., Strauss, J. & Yang, G. Cohort profile: The China Health and Retirement Longitudinal Study (CHARLS). *Int. J. Epidemiol.* **43**, 61–68 (2014).
18. QIAO Yucheng. Physical activity level: Ranking, measurement and energy reckoning. *Sports Res. Educ.* **32** (3), 1–12 (2017).
19. Kurz, C. F., Krzywinski, M. & Altman, N. Propensity score matching. *Nat. Methods.* **21** (10), 1770–1772 (2024).
20. Shi, S. et al. Comparison of neoadjuvant chemoradiotherapy versus chemoradiotherapy plus immunotherapy for esophageal squamous cell carcinoma in a real-world multicenter cohort: A propensity score matching study. *Sci. Rep.* **14** (1), 24738 (2024).
21. Jowsey, T., Yen, L. & W, P. M. Time spent on health related activities associated with chronic illness: A scoping literature review. *BMC Public. Health.* **12**, 1044 (2012).
22. Warburton, D. E. R., Nicol, C. W. & Bredin, S. S. D. Health benefits of physical activity: The evidence. *CMAJ Can. Med. Assoc. J. J. Assoc. Med. Can.* **174**, 801–809 (2006).
23. McPhee, J. S. et al. Physical activity in older age: Perspectives for healthy ageing and frailty. *Biogerontology* **17**, 567–580 (2016).
24. Marques, A., Santos, T., Martins, J., Matos, M. G. D. & Valeiro, M. G. The association between physical activity and chronic diseases in European adults. *Eur. J. Sport Sci.* **18**, 140–149 (2018).
25. Dhalwani, N. N. et al. Long terms trends of multimorbidity and association with physical activity in older English population. *Int. J. Behav. Nutr. Phys. Act.* **13**, 8 (2016).
26. Qiu, S. et al. Exercise training and endothelial function in patients with type 2 diabetes: A meta-analysis. *Cardiovasc. Diabetol.* **17**, 64 (2018).
27. Fuller, K. N. Z., Summers, C. M. & Valentine, R. J. Effect of a single bout of aerobic exercise on high-fat meal-induced inflammation. *Metabolism* **71**, 144–152 (2017).
28. Gleeson, M. et al. The anti-inflammatory effects of exercise: Mechanisms and implications for the prevention and treatment of disease. *Nat. Rev. Immunol.* **11**, 607–615 (2011).
29. Linke, A. et al. Antioxidative effects of exercise training in patients with chronic heart failure: Increase in radical scavenger enzyme activity in skeletal muscle. *Circulation* **111**, 1763–1770 (2005).
30. Firth, J. et al. Motivating factors and barriers towards exercise in severe mental illness: A systematic review and meta-analysis. *Psychol. Med.* **46**, 2869–2881 (2016).
31. Vr, V., Yf, C., Gc, H., Ej, T. & Mc, C. Low-intensity daily walking activity is associated with hippocampal volume in older adults. *Hippocampus* [Internet]. **25** (2015). https://pubmed.ncbi.nlm.nih.gov/25483019/(. Accessed 23 May 2024.
32. Sayed, N. et al. An inflammatory aging clock (iAge) based on deep learning tracks multimorbidity, immunosenescence, frailty and cardiovascular aging. *Nat. Aging.* **1**, 598–615 (2021).
33. Lin, Y. et al. Association between sleep and multimorbidity in Chinese elderly: Results from the Chinese longitudinal healthy longevity survey (CLHLS). *Sleep. Med.* **98**, 1–8 (2022).
34. Kyprianidou, M., Panagiotakos, D., Kambanaros, M., Makris, K. C. & Christophi, C. A. Quality of sleep in the Cypriot population and its association with multimorbidity: A cross-sectional study. *Front. Public. Health.* **9**, 693332 (2021).
35. Fabbri, L. M. et al. COPD and multimorbidity: Recognising and addressing a syndemic occurrence. *Lancet Respir Med.* **11**, 1020–1034 (2023).
36. Faria, A. C. et al. Drinking to death: Hyponatraemia induced by synthetic phenethylamines. *Drug Alcohol Depend.* **212**, 108045 (2020).
37. Masip, J. & Germà Lluch, J. R. Alcohol, health and cardiovascular disease. *Rev. Clin. Esp.* **221**, 359–368 (2021).

Author contributions

Bingbing FAN, Kexin Ren proposed the main research objectives and was responsible for the conception, design and implementation of the study; Bingbing FAN carried out the collection and organization of data, statistical processing, and manuscript writing; Lang Li was responsible for the drawing and presentation of figures and tables; Kexin Ren was in charge of quality control and reviewing the article, performed thesis revisions, took responsibility for the entirety of the article.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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